

- Note:**
1. Question paper consists of two parts (Part A and Part B)
 2. Answer all sub questions from **Part A**
 3. Answer all questions from **Part B** with **either or choice**

PART – A			10 X 2 = 20 M		
				BTL	Marks
I.	a)	Give the conditions for producing sustained interference fringes.	CO1	L3	2 M
	b)	Define resolving power of a diffraction grating.	CO1	L1	2 M
	c)	What are dielectric materials? Give examples.	CO2	L1	2 M
	d)	Define the terms magnetic field intensity, magnetic flux density.	CO2	L1	2 M
	e)	Write about coordination number and packing factor.	CO3	L1	2 M
	f)	Sketch the planes (100), (111) in a cubic crystal system.	CO3	L4	2 M
	g)	State Heisenberg’s uncertainty principle.	CO4	L1	2 M
	h)	Compute the deBroglie wavelength of an electron accelerated through a potential of 100 volts.	CO4	L5	2 M
	i)	What is Einstein’s equation in semiconductors.	CO5	L3	2 M
	j)	Write the expressions for drift and diffusion currents of a semiconductor.	CO5	L1	2 M
PART – B			5 X 10 = 50 M		
UNIT – I					
1.	a)	Explain how Newton’s rings are formed and derive an expression for the wavelength of light used.	CO1	L5	5 M
	b)	In Newton’s rings experiment, the diameter of the 8 th ring was 0.35 cm and diameter of 18 th ring was 0.65 cm. If the wavelength of light is 6000 Å ⁰ find the radius of curvature of the lens.	CO1	L3	5 M
(OR)					
2.	a)	Discuss Fraunhofer diffraction due to n slits and analyze diffraction grating.	CO1	L2	5 M
	b)	Discuss the phenomenon of double refraction.	CO1	L1	5 M
UNIT – II					
3.	a)	Deduce Claussius Mossotti equation in dielectrics.	CO2	L4	5 M
	b)	Analyze the variation of polarizability of a dielectric as a function of temperature.	CO2	L4	5 M
(OR)					
4.	a)	Discuss the origin of magnetic moment in an atom and derive an expression for orbital magnetic moment.	CO2	L6	5 M
	b)	Explain the variation of susceptibility with temperature in dia, para, ferro, antiferro and ferri magnetic materials.	CO2	L2	5 M
UNIT – III					
5.	a)	Explain the terms space lattice, basis, unit cell in crystalline solids.	CO3	L2	5 M
	b)	Describe the seven crystal systems.	CO3	L4	5 M
(OR)					
6.	a)	Describe working of Bragg’s X-ray diffractometer.	CO3	L2	5 M
	b)	Explain how Laue method is used to determine crystal structure.	CO3	L2	5 M
UNIT – IV					
7.	a)	Mention the properties of matter waves.	CO4	L2	5 M
	b)	Derive Schrodinger’s time independent wave equation.	CO4	L2	5 M
(OR)					

8.	a)	List out the failures of Classical free electron theory and explain the salient features of Quantum free electron theory.	CO4	L2	10M
UNIT – V					
9.	a)	Distinguish between intrinsic and extrinsic semiconductors.	CO5	L3	5 M
	b)	State and explain Hall effect. Derive Hall Coefficient.	CO5	L4	5 M
(OR)					
10	a)	Show that Fermi energy level is in the middle of valence band and conduction band in an intrinsic semiconductor.	CO5	L3	
	b)	Derive an expression for carrier concentration of an intrinsic semiconductor.	CO5	L5	5 M

***** End of the Question Paper *****